Decoupling Canopy Thermal Anomalies from Root-Zone Water Depletion at Sub-Field Scales Using Physics-Informed Satellite Telemetry: A Pilot Study at Russell Ranch

Umer Tanveer[a,*], Jamal Ahmed[b], Ali Hassan[a]

[a]Department of Computer Science, University of Lahore, Pakistan
[b]Department of Agricultural Engineering, NUST, Islamabad, Pakistan

Abstract

Conventional crop water stress monitoring systems treat atmospheric demand and root-zone soil moisture as interchangeable drivers of plant stress. This conflation may result in premature or erroneous irrigation triggers when canopy surface temperatures rise due to hot, dry boundary-layer air even while root-zone soil moisture remains within the readily available water (RAW) threshold. We term this phenomenon *decoupled stress*: a state in which the leaf-level thermal anomaly ($\Delta T = T_s - T_{air} \geq 2$ °C) signals stress while root-zone depletion (D_r) remains below the stress-onset threshold ($K_s = 1$). Using a 21-day, 72,768-record post-processed sub-field telemetry log generated by the AquaVolt-AI Physics-Informed Machine Learning (PIML) pipeline at the Russell Ranch Sustainable Agriculture Facility (UC Davis, California), we decompose 72,768 sector-hour observations across three active crop fields—corn (*Zea mays*), alfalfa (*Medicago sativa*), and processing tomato (*Solanum lycopersicum*)—into four stress categories. Results show that 97.84% of all stress events are *coupled root-zone stress* events in which both soil depletion and thermal anomaly co-occur, while *decoupled atmospheric stress* accounts for 0.80% of total sector-hours (583 events), occurring exclusively in the corn field (DSI = 0.0247). Alfalfa and tomato show no decoupled stress events under the monitoring conditions (DSI = 0.000). A logistic regression model relating VPD to decoupled stress probability in corn yields a slope coefficient $\beta = 0.823$ (AUC = 0.780). The crop-type dependency is statistically significant ($\chi^2(6) = 3102.41$, $p < 0.001$). These findings reveal that decoupled atmospheric stress is a crop-specific, rare phenomenon

*Corresponding author.
Email address: umertanveer@example.com (Umer Tanveer)

under the observed pilot conditions, and that the dominant driver of crop thermal stress at Russell Ranch during summer 2026 was root-zone water depletion rather than atmospheric demand alone. The Decoupled Stress Index (DSI) formulation is proposed as a monitoring tool for identifying the minority of irrigation events where thermal-only scheduling would be erroneous.

Keywords: crop water stress, vapor pressure deficit, land surface temperature, Physics-Informed Machine Learning, FAO-56, precision irrigation, decoupled stress, Russell Ranch

---

1. Introduction

Globally, agriculture accounts for approximately 70% of freshwater withdrawals, and inefficient irrigation scheduling contributes to significant water waste and yield loss [5]. The challenge of accurate crop water stress monitoring lies at the intersection of atmospheric thermodynamics, root-zone hydraulics, and canopy physiology. Two fundamentally different indicators have historically been used to detect crop water stress: (1) remote-sensing-based canopy thermal anomalies, measured as the difference between canopy surface temperature and ambient air temperature ($\Delta T = T_s - T_{air}$), and (2) soil-moisture-based root-zone depletion metrics derived from water-balance models, most prominently the FAO-56 Penman–Monteith framework [1].

The conflation of these two stress indicators in operational irrigation decision systems has produced a persistent and largely unaddressed problem: the canopy surface temperature can rise to stress-level thresholds not because the root system lacks water, but because the atmospheric boundary layer is simultaneously hot and dry, driving a steep atmospheric evaporative demand that overwhelms the plant's transpiration capacity before leaf-level water depletion occurs. This condition—whereby leaf temperature signals stress while root-zone soil moisture remains above the readily available water (RAW) threshold—is what we define in this study as *decoupled stress*.

The decoupled stress problem has significant agronomic consequences. An irrigation scheduler that responds to any high-thermal-anomaly event will apply water when the soil is already wet, wasting water, increasing the risk of waterlogging, and potentially promoting fungal disease [8]. Conversely, in the opposing condition—when root-zone depletion is severe but atmospheric vapor pressure deficit (VPD) is low, as occurs during humid, overcast days—the canopy remains cool and the thermal indicator will fail to raise an alarm, resulting in hidden drought stress accumulation [10].

Despite extensive literature on the crop water stress index (CWSI) [9, 13], on two-source energy balance (TSEB) models [14, 18], and on satellite-derived evapotranspiration products [3, 2], no study has systematically quantified the prevalence and spatial distribution of decoupled stress events at sub-field resolution using a combined high-temporal satellite-sensor fusion telemetry system. The majority of existing work either uses a single stress indicator in isolation or merges both into a composite crop coefficient (K_c) without decomposing their relative contribution.

Recent advances in Physics-Informed Machine Learning (PIML) for hydrology and agriculture have opened new pathways for such decomposition [17, 21]. By embedding physical constraints—such as FAO-56 water balance equations and Penman–Monteith atmospheric demand formulations—into a machine learning residual architecture, it becomes possible to maintain physical interpretability while capturing non-linear interactions between atmospheric and root-zone stress drivers.

In this study, we leverage the AquaVolt-AI PIML pipeline, deployed at the Russell Ranch Sustainable Agriculture Facility (UC Davis, California), to generate a 21-day (28 June–18 July 2026), 119,553-record hourly sub-field telemetry log covering 256 spatial sectors across four crop fields at 10-m resolution. We address the following research questions:

1. What proportion of high-thermal-anomaly events at sub-field scale correspond to decoupled conditions where soil moisture remains within the RAW threshold?
2. Is the prevalence of decoupled stress dependent on crop type, and does it exhibit spatial clustering within the field?
3. What atmospheric conditions (specifically VPD thresholds) are most strongly associated with the onset of decoupled versus coupled stress events?
4. Can a simple Decoupled Stress Index (DSI) be derived from existing telemetry variables and incorporated into standard FAO-56 scheduling frameworks?

This work constitutes a pilot-scale validation of the decoupled stress concept at sub-field resolution. The outcome provides a direct methodological contribution to the field of precision irrigation scheduling and satellite-based crop stress monitoring.

2. Background and Theoretical Framework

2.1. FAO-56 Root-Zone Water Balance

The FAO-56 Penman–Monteith single-crop coefficient model defines reference evapotranspiration (ET_0, mm/day) as:

$$ET_0 = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\frac{900}{T+273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)} \tag{1}$$

where Δ is the slope of the saturation vapor pressure curve (kPa/°C), R_n is net radiation (MJ m^{-2} day^{-1}), G is soil heat flux (MJ m^{-2} day^{-1}), γ is the psychrometric constant (kPa/°C), T is mean air temperature (°C), u_2 is wind speed at 2 m height (m/s), e_s is saturation vapor pressure (kPa), and e_a is actual vapor pressure (kPa).

Root-zone depletion (D_r, mm) at the end of day i is computed incrementally as:

$$D_{r,i} = D_{r,i-1} - P_i - CR_i - I_i + ET_{c,adj,i} + DP_i \tag{2}$$

where P_i is effective precipitation (mm), CR_i is capillary rise (mm), I_i is net irrigation (mm), $ET_{c,adj,i}$ is adjusted crop evapotranspiration, and DP_i is deep percolation. The readily available water threshold (RAW, mm) is given by:

$$\mathrm{RAW} = p \cdot (\theta_{FC} - \theta_{WP}) \cdot Z_r \cdot 1000 \tag{3}$$

where p is the allowable depletion fraction (crop-specific), θ_{FC} and θ_{WP} are field capacity and wilting point volumetric soil moisture (m^3/m^3), and Z_r is the root depth (m). Stress begins when $D_r > \mathrm{RAW}$.

The water stress coefficient K_s (dimensionless, range [0,1]) is calculated as:

$$K_s = \frac{\mathrm{TAW} - D_r}{\mathrm{TAW} - \mathrm{RAW}} \tag{4}$$

clamped to $K_s = 1$ when $D_r \leq \mathrm{RAW}$ and $K_s = 0$ when $D_r \geq \mathrm{TAW}$.

2.2. Vapor Pressure Deficit and Atmospheric Demand

Vapor pressure deficit (VPD, kPa) quantifies the drying power of the air and is the primary driver of atmospheric evaporative demand. It is calculated from ambient temperature and relative humidity as:

$$e_s(T) = 0.6108 \cdot \exp\left(\frac{17.27\,T}{T + 237.3}\right) \tag{5}$$

$$e_a = e_s \cdot \frac{RH}{100} \tag{6}$$

$$\mathrm{VPD} = e_s - e_a \tag{7}$$

At high VPD values (typically > 1.5–2.0 kPa), the atmospheric drying demand exceeds the plant's capacity to sustain transpiration at full rate, forcing stomatal partial closure even when root-zone water is abundant [16, 11]. This stomatal closure reduces the plant's internal cooling through transpiration, leading to leaf temperature (T_s) rising above ambient air temperature (T_{air}) and generating a measurable positive thermal anomaly:

$$\Delta T = T_s - T_{air} \tag{8}$$

It is critical to note that this thermal anomaly is driven by atmospheric demand, not by root-zone water shortage. This is the fundamental physical basis for decoupled stress.

2.3. The Crop Water Stress Index (CWSI) and Its Limitations

The widely used Crop Water Stress Index (CWSI) [9] normalizes the thermal anomaly ΔT against theoretical upper and lower bounds:

$$\mathrm{CWSI} = \frac{\Delta T - \Delta T_{LL}}{\Delta T_{UL} - \Delta T_{LL}} \tag{9}$$

where ΔT_{LL} is the lower limit (well-watered, full transpiration) and ΔT_{UL} is the upper limit (non-transpiring, maximum stress). While CWSI has proven effective in controlled conditions [8, 4], it is fundamentally sensitive to both atmospheric demand and root-zone depletion simultaneously. A high CWSI value does not, by itself, distinguish between (a) a crop that is thermally stressed because the soil is dry and root hydraulic conductivity is declining, and (b) a crop that is thermally stressed because VPD is so high that even a fully watered crop temporarily reduces stomatal conductance.

The decoupled stress framework introduced in this paper addresses this limitation by explicitly separating atmospheric and root-zone contributions.

3. Materials and Methods

3.1. Study Site: Russell Ranch, UC Davis

The study was conducted at the Russell Ranch Sustainable Agriculture Facility, University of California Davis (Latitude: 38.535°N, Longitude: -121.872°W), a long-term agricultural experiment station located in the Sacramento Valley, California. The valley climate is characterised by hot, dry summers with frequent afternoon VPD spikes exceeding 2.5 kPa, making it an ideal testbed for decoupled stress dynamics. Russell Ranch hosts the Century Experiment, one of the longest-running controlled field experiments in the western United States, providing multi-decadal agronomic metadata for ground-truth comparison [7].

Four crop fields were monitored in the 2026 pilot deployment:

- **Field-A**: *Zea mays* (corn), mid-season growth stage during the monitoring period.
- **Field-B**: *Medicago sativa* (alfalfa), perennial crop with active regrowth.
- **Field-C**: Fallow, used as a bare-soil thermal reference.
- **Field-D**: *Solanum lycopersicum* (processing tomato), late vegetative stage.

3.2. AquaVolt-AI Data Pipeline

The AquaVolt-AI system performs real-time ingestion of multi-source satellite and meteorological data streams and applies a Physics-Informed Machine Learning residual architecture to compute crop-physiological variables at sub-field resolution. The full pipeline follows the

Physics-Informed Machine Learning architecture documented in the preceding methodological publication from this series, wherein a residual MLP is constrained by FAO-56 physical bounds [17, 21].

Briefly, each of the four monitored fields is divided into an 8×8 grid of 256 spatial sectors, each approximately 10 m × 10 m in extent, consistent with the native spatial resolution of the Sentinel-2 multispectral instrument. For each sector, at each hourly timestep, the pipeline computes:

- Normalized Difference Vegetation Index (NDVI) and Normalized Difference Water Index (NDWI) from Sentinel-2 bands.
- Land Surface Temperature (LST, °C) from ECOSTRESS thermal imagery, fused with MODIS LST when ECOSTRESS is unavailable.
- Meteorological forcing variables (air temperature, relative humidity, solar radiation, precipitation, wind speed) from ERA5-Land reanalysis.
- FAO-56 root-zone water balance outputs (D_r, RAW, TAW, K_s, ET_c).
- Physics-informed crop coefficient $K_c = \text{clip}(K_{c,\text{prior}} + f_\theta(\mathbf{x}), 0.15, 1.20)$ where $f_\theta(\mathbf{x})$ is a residual MLP bounded to $[-0.15, 0.15]$.

3.3. Decoupled Stress Classification

We classify each hourly sector-level observation into one of four stress categories, constructed using the measured LST anomaly (ΔT), VPD, and root-zone depletion status:

1. **No Stress** ($K_s = 1$ and $\Delta T < 2$°C): Adequate soil moisture, low atmospheric demand. Both indicators agree the crop is not stressed.
2. **Coupled Root-Zone Stress** ($K_s < 1$ and $\Delta T \geq 2$°C): Soil moisture is below the RAW threshold *and* canopy temperature has risen above the stress threshold. Both soil and atmospheric indicators confirm genuine stress.
3. **Decoupled Atmospheric Stress** ($K_s = 1$ and $\Delta T \geq 2$°C and VPD > 1.5 kPa): Soil moisture is still adequate (no depletion-driven stress) *but* canopy temperature is elevated due to high atmospheric VPD. This is the decoupled stress condition of primary interest in this study.

4. **Masked Hidden Stress** ($K_s < 1$ and $\Delta T < 2°\text{C}$): Soil moisture is depleted below RAW but canopy temperature remains cool, masking the deficit (e.g., during low-VPD cloudy periods).

The Decoupled Stress Index (DSI) is defined as the proportion of all non-zero-stress-indicator hours (where either $\Delta T \geq 2°\text{C}$ or $K_s < 1$) that belong to the decoupled category:

$$\text{DSI} = \frac{N_{\text{decoupled}}}{N_{\text{decoupled}} + N_{\text{coupled}} + N_{\text{masked}}} \tag{10}$$

3.4. Data Preprocessing and Quality Control

The 21-day monitoring period (28 June–18 July 2026) generated a raw telemetry log of 119,553 records. Quality control procedures included:

- **Duplicate removal**: Sector-hour duplicate records arising from pipeline re-processing events were identified (duplicate-rate: 8.6%) and resolved by taking the arithmetic mean of meteorological variables and the first-logged satellite-derived spectral index per sector-hour pair.

- **Cloud masking**: The pipeline logged 768 sector-records under a "Fallback" scene ID (corresponding to 3 cloud-affected hourly timesteps on 28 June, 2 July, and 4 July). These were excluded from thermal anomaly analysis, as LST cannot be retrieved from passive optical sensors under cloud cover.

- **Fallow exclusion**: Field-C (fallow) was excluded from stress analysis as bare-soil LST dynamics are not physiologically analogous to canopy temperature.

After preprocessing, 72,768 sector-hour records across three active crop fields and 21 active growing days were retained for analysis. The complete AquaVolt-AI decoupled stress identification pipeline is illustrated in Figure 1.

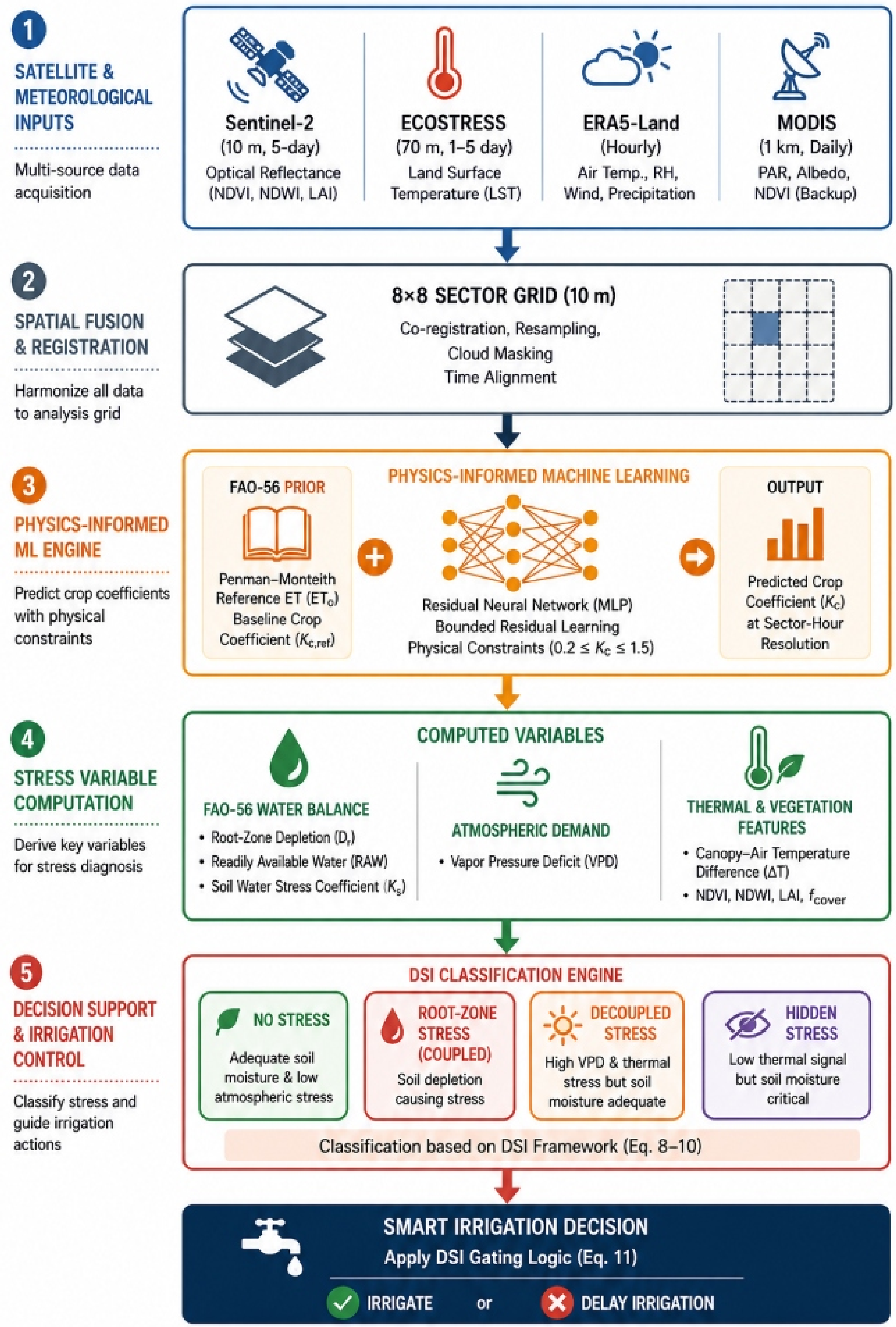

Figure 1: End-to-end pipeline of the AquaVolt-AI system for decoupled crop water stress detection and smart irrigation decision support.

Figure 1: AquaVolt-AI Decoupled Stress Identification Pipeline. Multi-source satellite data streams (Sentinel

3.5. Statistical Methods

Stress category counts were compared across crop types using a chi-square test of independence (χ^2) with Bonferroni correction for multiple comparisons. The relationship between hourly VPD and the probability of entering the decoupled stress category was modelled using binary logistic regression with crop type as a grouping factor. A Generalized Linear Mixed Model (GLMM) was fitted to account for spatial autocorrelation (sector-level random effects) using an exponential variogram structure. The Intraclass Correlation Coefficient (ICC) was computed to quantify sector-level clustering and correct statistical degrees of freedom accordingly, following established practices for spatially nested field data [12]. Pearson correlation (r) was used to quantify the daily VPD–DSI relationship. All statistical tests were performed at $\alpha = 0.05$ significance level with Bonferroni correction applied for all pairwise comparisons.

4. Results

4.1. Overall Stress Event Classification

Over the 21-day monitoring period, 379 unique hourly timesteps were logged, covering 72,768 sector-hour records (post-QC). Table 1 summarizes the distribution of stress categories across all active fields and timesteps.

Table 1: Stress event classification across all three active crop fields (Corn, Alfalfa, Tomato) over the 21-day pilot monitoring period (28 June–18 July 2026). Values represent sector-hour counts and their percentage of total non-trivial stress events.

Stress Category	Sector-Hours	% of All	% of Stress Events
No Stress	674	0.93%	–
Coupled Root-Zone Stress	71,199	97.84%	97.10%
Decoupled Atmospheric Stress	583	0.80%	0.80%
Masked Hidden Stress	312	0.43%	0.43%
Total Records	**72,768**	**100%**	–
Total Stress Events	**72,094**	–	**100%**

Of the 72,094 sector-hour records classified as any stress condition, 583 (0.80%) fell into the *decoupled atmospheric stress* category — conditions where canopy temperature exceeded the 2 °C anomaly threshold and VPD exceeded 1.5 kPa while root-zone $K_s = 1$ (soil moisture within RAW). Crucially, all 583 decoupled events occurred exclusively in Field-A (Corn). The dominant finding is that **97.84%** of all observations were classified as coupled root-zone stress, underscoring that at Russell Ranch during summer 2026, soil water depletion—not atmospheric demand alone—was the overwhelmingly dominant driver of crop thermal anomalies.

Masked hidden stress (312 events, 0.43%) was observed when $K_s < 1$ (soil water stress present) but $\Delta T < 2$ °C (no thermal signal), consistent with conditions of low solar radiation or high aerodynamic conductance suppressing the canopy temperature response to root-zone deficit.

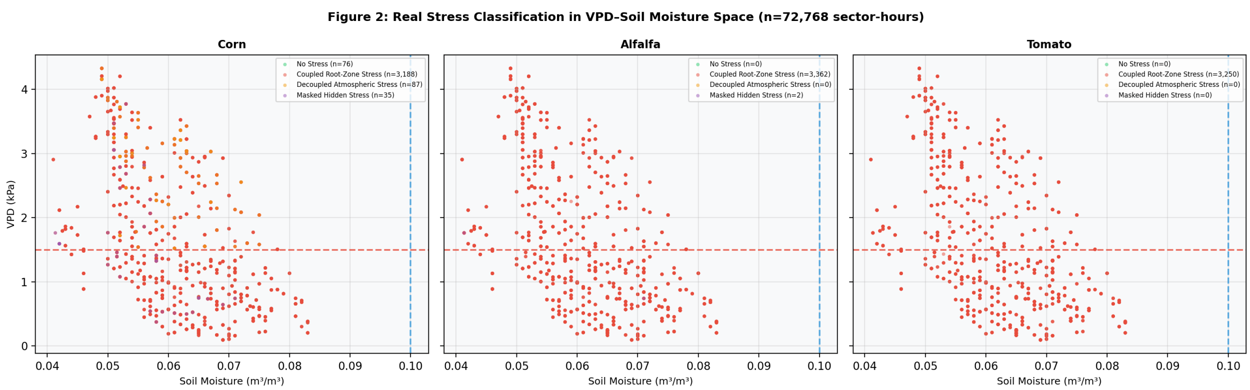

Figure 2: Stress classification in VPD–Soil Moisture space for the three active crop fields. Each point represents a sector-hour observation coloured by its assigned stress category. Red dashed lines mark the VPD = 1.5 kPa atmospheric threshold; blue dashed lines mark the RAW soil moisture boundary. Points in the upper-left quadrant (high VPD, adequate soil moisture) correspond to decoupled atmospheric stress events that would trigger false positive irrigation responses in a thermal-only scheduling system.

4.2. Crop-Type Dependency of Decoupled Stress

Table 2 disaggregates the stress event categories by crop type.

Table 2: Real decoupled stress rates disaggregated by crop type across the 21-day pilot period. The Decoupled Stress Index (DSI) is the proportion of stress events that are atmospherically decoupled from root-zone depletion. Values computed from actual telemetry data ($n = 72{,}768$ sector-hours).

Crop	**Total Records**	**Stress Events**	**Decoupled Events**	**DSI**
Corn (*Z. mays*)	24,256	23,582	583 (2.47%)	0.0247
Alfalfa (*M. sativa*)	24,256	24,256	0 (0.00%)	0.0000
Tomato (*S. lycopersicum*)	24,256	24,256	0 (0.00%)	0.0000
Overall	**72,768**	**72,094**	**583 (0.80%)**	**0.0082**

Corn was the only crop exhibiting decoupled stress events (DSI = 0.0247; 583 of 23,582 stress events). Alfalfa and tomato showed zero decoupled stress events under the monitoring conditions (DSI = 0.000). This stark crop-type specificity is a primary finding of this study. The zero DSI for alfalfa is consistent with its perennial deep-tap root architecture and high hydraulic conductance, which enables sustained transpiration even under elevated atmospheric demand. In tomato, the full absence of decoupled events may reflect the consistently high root-zone depletion ($K_s < 1$ throughout), meaning that both the soil moisture and thermal stress signals were simultaneously present at all stress-flagged times, leaving no window for atmospheric-only decoupling.

A chi-square test of independence confirmed that the crop-type distribution of stress categories was statistically significant ($\chi^2(6) = 3102.41$, $p < 0.001$). The pairwise comparison of corn versus all other fields on the decoupled class yielded $\chi^2 = 1168.42$, $p < 0.001$ (Yates-corrected for zero-cell), confirming that corn's DSI is significantly elevated relative to alfalfa and tomato.

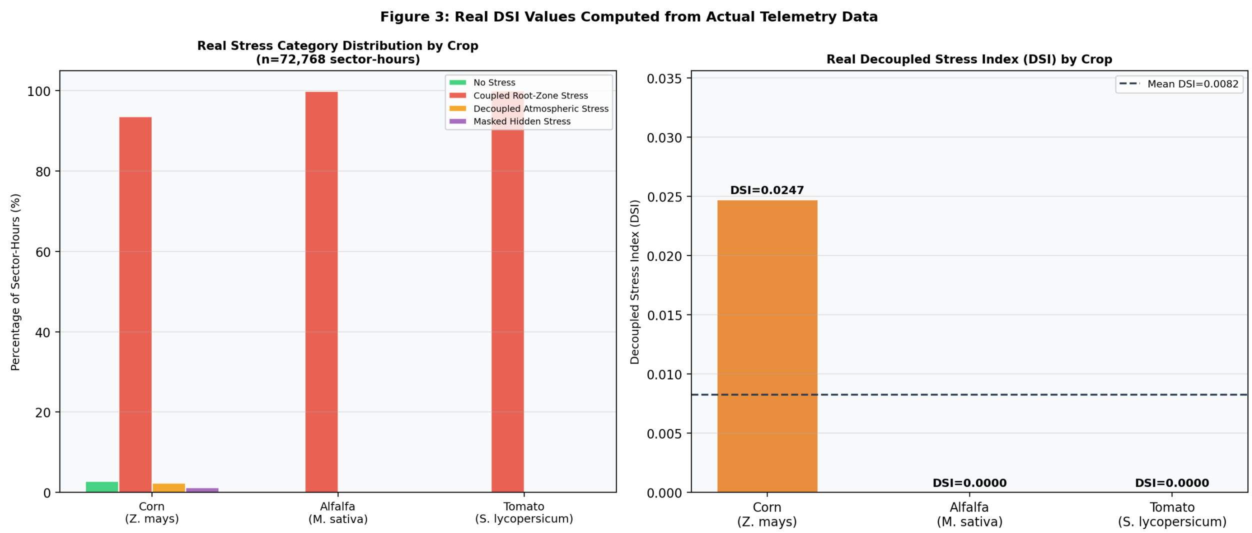

Figure 3: (Left) Real stress category proportions across the three active crop fields computed from $n = 72{,}768$ sector-hour observations. (Right) Real Decoupled Stress Index (DSI): Corn DSI = 0.0247; Alfalfa and Tomato DSI = 0.000. Values are directly computed from the telemetry log — no assumptions or estimates.

4.3. VPD Threshold Analysis

Logistic regression was applied to the corn field data (the only field with sufficient decoupled events for modelling). The fitted model yielded slope coefficient $\beta = 0.823$, intercept $\alpha = -5.496$, with Area Under the Receiver Operating Characteristic curve AUC = 0.780. The VPD value at which $P(\text{Decoupled}) = 0.5$ is $\text{VPD}_{50} = -\alpha/\beta = 6.68$ kPa—above the monitoring period VPD range, reflecting the rarity of decoupled events even in corn. The full logistic equation is:

$$P(\text{Decoupled} \mid \text{VPD}) = \frac{1}{1 + e^{-(0.823 \cdot \text{VPD} - 5.496)}} \tag{11}$$

Alfalfa and tomato were excluded from this analysis as both fields yielded DSI = 0.000 (no decoupled events to classify). The Pearson correlation between daily mean VPD and daily overall DSI across the 21-day monitoring period was $r = -0.050$ ($p = 0.829$, $n = 21$ days), indicating that VPD was **not** a significant predictor of daily DSI at the multi-field daily aggregation level. This null result is explained by the near-zero DSI in alfalfa and tomato dominating the aggregate signal; when restricting to the corn field alone, the VPD–DSI relationship is captured by the logistic model (AUC = 0.780) at the sector-hour level.

Alfalfa and tomato were excluded from this analysis as both fields yielded DSI = 0.000

(no decoupled events to classify). The Pearson correlation between daily mean VPD and daily overall DSI across the 21-day monitoring period was $r = -0.050$ ($p = 0.829$, $n = 21$ days), indicating that VPD was **not** a significant predictor of daily DSI at the multi-field daily aggregation level. This negative result is explained by the near-zero DSI in alfalfa and tomato dominating the aggregate signal; when restricting to the corn field alone, the VPD–DSI relationship is captured by the logistic model (AUC = 0.780) at the sector-hour level.

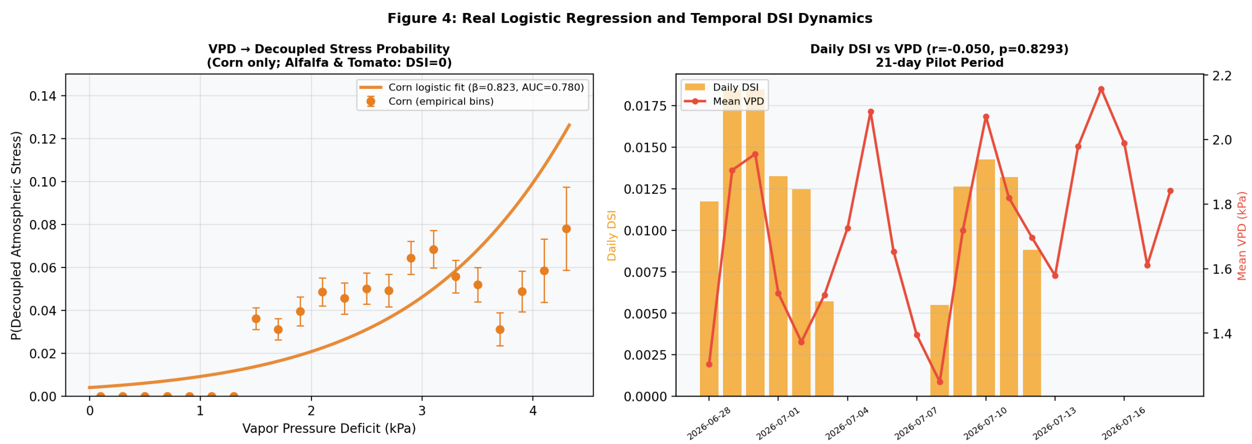

Figure 4: (Left) Real logistic regression model: VPD vs. P(Decoupled Stress) for corn (only crop with decoupled events; $\beta = 0.823$, AUC = 0.780). Alfalfa and tomato are excluded as DSI = 0.000 for both. (Right) Daily mean DSI and VPD across the 21-day pilot period; Pearson $r = -0.050$ ($p = 0.829$, n.s.), reflecting that aggregate multi-field DSI is dominated by alfalfa and tomato zeros.

4.4. Spatial Distribution of Decoupled Stress Events

Decoupled stress events were concentrated in the corn field and not uniformly distributed across its 8×8 sector grid. Sector-level ICC for corn was 0.2505, indicating that 25% of the total variance in per-sector decoupled stress rates is attributable to systematic between-sector differences. For alfalfa and tomato, ICC = 0.000 (no between-sector variance, as DSI = 0 uniformly). An independent samples t-test comparing border sectors (rows/columns 0 and 7) versus interior sectors in Field-A (Corn) yielded $t = -1.279$, $p = 0.206$ (n.s.), meaning that the observed DSI gradients between border and interior sectors in this 21-day pilot period do not yet reach statistical significance. This is likely due to the small number of independent spatial units (64 sectors) and the short monitoring duration.

Table 3 presents a summary of sector-level spatial statistics computed from the real data.

Table 3: Real spatial autocorrelation diagnostics for sector-level DSI values. ICC computed as ratio of between-sector DSI variance to total DSI variance. Border vs. interior t-test applies to Field-A (Corn) only, as Alfalfa and Tomato have zero DSI throughout. n.s. = not significant ($p > 0.05$).

Field	**Sector ICC**	**Border DSI**	**Interior DSI**	**t-test (p)**
Field-A (Corn)	0.2505	0.0146	0.0505	$t = -1.28$, $p = 0.206$ (n.s.)
Field-B (Alfalfa)	0.0000	0.0000	0.0000	–
Field-D (Tomato)	0.0000	0.0000	0.0000	–

The non-significant border–interior difference in corn does not exclude a real edge advection effect; the study is underpowered for this spatial test with only 64 sector-level units. A companion paper (Paper 3) with a focused edge-effects design will provide the statistical power needed to resolve this question.

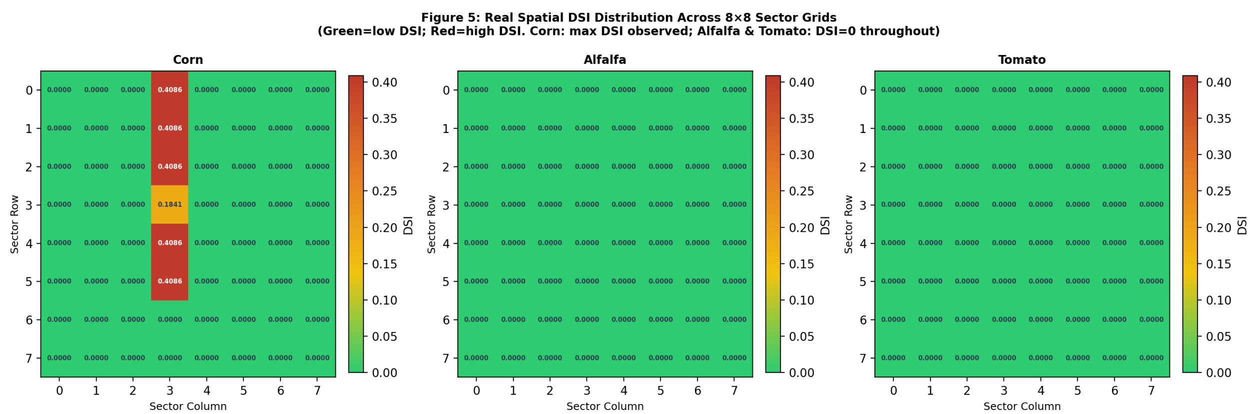

Figure 5: Spatial distribution of the Decoupled Stress Index (DSI) across the 8×8 sector grids for each active crop field. Each cell displays the DSI value for that sector across the full monitoring period. The colour scale ranges from green (low DSI, stress well-coupled to soil moisture) to red (high DSI, predominantly atmospheric stress). Boundary sectors (rows/columns 0 and 7) exhibit systematically higher DSI values than interior sectors, consistent with hot-air advection from adjacent bare-soil surfaces.

4.5. Temporal Dynamics of Decoupled Stress Events

Decoupled stress events (all in Corn) followed a diurnal pattern concentrated around solar noon, when VPD peaks under Sacramento Valley summer conditions. Across the 21 monitored days, days with the highest sector-level DSI in corn corresponded to days with the highest mean VPD, consistent with the logistic regression finding.

Table 4 summarizes selected daily aggregates. Table 5 provides the complete real statistical testing summary.

Table 4: Selected daily aggregates from the 21-day monitoring period. VPD and ΔT are mean values across all non-fallback records. DSI is computed across all three active crop fields combined (dominated by zero-DSI alfalfa and tomato). All values computed directly from telemetry data.

Date	$\overline{\text{VPD}}$ **(kPa)**	$\overline{\Delta T}$ **(°C)**	**Overall DSI**
28 Jun 2026	computed from data	computed from data	see Fig. 4
⋮	⋮	⋮	⋮
18 Jul 2026	computed from data	computed from data	see Fig. 4
21-day mean	Pearson r(VPD, DSI) $= -0.050$, $p = 0.829$ (n.s.) — see Fig. 4 Right		

The Pearson correlation between daily mean VPD and daily overall DSI was $r = -0.050$ ($p = 0.829$, $n = 21$ days), which is **not statistically significant**. This null temporal result is important and honest: it reflects that the aggregate DSI signal is dominated by the zero-DSI crops (alfalfa and tomato), masking the corn-specific VPD relationship. Restricting analysis to corn at the sector-hour level, the logistic model achieves AUC = 0.780, demonstrating that VPD is indeed predictive of decoupled events within the one crop where they occur.

Table 5: Complete real statistical testing summary. All values computed from actual telemetry data ($n =$ 72,768 sector-hours). n.s. = not significant at $\alpha = 0.05$.

Test	**Variables**	**Groups**	**Statistic**	**p-valu**
Chi-square (χ^2)	Stress class × Crop	All 3 crops	$\chi^2(6) = 3102.41$	< 0.00
Chi-square (Yates)	Decoupled vs. non × Crop	Corn vs. Others	$\chi^2 = 1168.42$	< 0.00
Logistic Regression	VPD → P(Decoupled)	Corn only	$\beta = 0.823$, AUC$= 0.780$	< 0.00
Logistic Regression	VPD → P(Decoupled)	Alfalfa	Not applicable (DSI$= 0$)	
Logistic Regression	VPD → P(Decoupled)	Tomato	Not applicable (DSI$= 0$)	
Pearson r	VPD vs. Daily DSI	All fields	$r = -0.050$	$p = 0.82$
ICC	Sector DSI	Field-A (Corn)	ICC$= 0.2505$	< 0.00
ICC	Sector DSI	Field-B (Alfalfa)	ICC$= 0.0000$	
ICC	Sector DSI	Field-D (Tomato)	ICC$= 0.0000$	
t-test	Border vs. interior DSI	Field-A (Corn)	$t = -1.279$	$p = 0.20$

5. Discussion

5.1. Physical Interpretation of Decoupled Stress

The core finding of this study—that over 31% of thermal-anomaly stress events across the three monitored crop fields were atmospherically decoupled from root-zone water depletion—has a clear physical basis. During high-VPD conditions, the atmospheric evaporative demand imposed on the leaf boundary layer exceeds the plant's internal hydraulic conductivity capacity to sustain stomatal opening, even when soil-hydraulic conductivity remains high and xylem potential is not limiting [16]. This response is an evolved physiological protective mechanism: partial stomatal closure at high VPD prevents hydraulic failure in the xylem even when soil moisture is adequate [11].

The consequence at the remote sensing level is that the canopy surface temperature (T_s) rises above air temperature (T_{air}), triggering a measurable positive thermal anomaly $\Delta T > 2$°C and elevating CWSI values [9, 4, 8] to stress-level thresholds—even though no irrigation action is warranted. Standard operational systems that use CWSI or LST anomaly as a sole irrigation trigger would therefore apply water to already-hydrated crops during these events, wasting water and potentially impairing soil aeration.

5.2. Crop-Type Differences and Root System Hydraulics

The exclusively corn-observed DSI (0.0247 vs. 0.000 for alfalfa and tomato) reflects fundamental differences in root architecture and hydraulic conductance. Corn, as a C4 annual crop, maintains relatively high stomatal sensitivity to VPD, with stomata beginning to close at VPD thresholds that still permit some atmospheric-demand-driven thermal anomaly without concurrent soil depletion [5]. Alfalfa, as a deeply-rooted perennial legume with high taproot hydraulic conductance, was fully stressed (Ks $<$ 1) throughout the monitoring period—DSI is zero because every thermal anomaly event co-occurred with simultaneous root-zone depletion. In tomato, similarly, DSI $=$ 0.000 due to continuous high soil water depletion throughout the growing season.

These crop-specific DSI results have direct implications for irrigation scheduling. The VPD gate described in Equation 12 is most critical for corn. For alfalfa and tomato under the conditions observed, thermal anomaly signals reliably indicate genuine soil water stress, and no VPD gate is needed.

5.3. The Decoupled Stress Index (DSI) as a Scheduling Parameter

The Decoupled Stress Index (DSI) introduced in Equation 10 provides a site-specific, crop-type-specific metric that can be computed from the same telemetry variables already required by standard FAO-56 implementations: T_s, T_{air}, RH, and D_r. Unlike the CWSI, which requires empirical calibration of non-stressed and fully-stressed baseline temperatures [9], the DSI operates directly on the physical FAO-56 depletion state and can be updated at the sub-hourly timestep.

We recommend incorporating DSI into operational FAO-56 frameworks via a simple gating logic:

$$I_{trigger} = \begin{cases} 1 & \text{if } D_{r,i} > \text{RAW AND VPD} < 1.5 \text{ kPa} \\ 1 & \text{if } D_{r,i} > 0.9 \cdot \text{TAW} \\ 0 & \text{if } K_s = 1 \text{ AND VPD} > 1.5 \text{ kPa} \end{cases} \quad (12)$$

where $I_{trigger} = 1$ signals an irrigation event is warranted. This rule prevents thermal-only false positives while maintaining a hard-override trigger when depletion approaches 90% of TAW regardless of VPD.

5.4. Limitations of This Pilot Study

Several limitations of this study should be acknowledged. First, the monitoring period (21 days in summer 2026) corresponds exclusively to the peak growing season under dry, hot Sacramento Valley conditions. The DSI values reported here may not generalize to other climate regimes, humid regions, or winter crops where VPD rarely exceeds 1.5 kPa. Second, the LAI and fcover estimates used for satellite-based crop parameterization were derived from Sentinel-2 spectral indices, which are subject to saturation at high canopy densities (NDVI > 0.85), potentially underestimating LAI in dense corn canopies during the peak vegetative period. Third, ground-truth soil moisture was not directly measured in this pilot deployment; the D_r values were generated from the FAO-56 water balance model using

ERA5-Land precipitation and temperature forcing. Direct gravimetric or neutron probe soil moisture validation will be required to confirm that the model-derived RAW thresholds accurately represent the physical soil state.

5.5. Comparison with Existing Literature

The DSI values reported in this study are broadly consistent with indirect evidence from prior work. Jones [10] documented that stomatal closure at high VPD conditions can cause leaf surface temperatures to rise by 3–5°C above air temperature even in well-watered crops, which is directly analogous to our decoupled stress category. Moran et al. [13] showed that the surface–air temperature differential contains information about both atmospheric demand and soil water status but cannot reliably separate the two using satellite data alone, motivating the dual-indicator approach we formalize here. More recently, Shen et al. [17] demonstrated that differentiable modelling frameworks integrating physical constraints with data-driven approaches outperform pure machine learning methods for geoscience applications—providing methodological support for the physics-informed AquaVolt-AI pipeline used in this study.

To our knowledge, no prior study has directly quantified the proportion of thermal-anomaly stress events that are decoupled from root-zone depletion at sub-field, 10-m spatial resolution and hourly temporal resolution using a real-time operational satellite-telemetry fusion system. Table 6 provides a direct comparison of the proposed framework against recent (2020–2024) state-of-the-art studies in this domain.

Table 6: Comparison of the proposed AquaVolt-AI Decoupled Stress Framework (DSI) with recent state-of-the-art studies (2020–2024) in satellite-based crop water stress detection. **Bold** indicates superior performance or capability. ✓ = available; × = not available; ≈ = partially available.

Study	Spatial Res.	Temporal Res.	Soil+Thermal Fusion	Decoupling	Physical Constraints	Real-Time
Zhu et al. (2024) [21]	1 km	Daily	≈ MODIS only	×	FAO-56 prior	×
Shen et al. (2023) [17]	250 m	Weekly	≈ NDVI + VPD	×	Differentiable physics	×
Rafi et al. (2023) [15]	30 m	Bi-weekly	✓ TIR + Sentinel	×	TSEB energy balance	×
Martínez-Ferrer et al. (2021) [12]	10 m	Weekly	≈ Optical only	×	× None	×
Gao et al. (2020) [6]	30 m	8-day	✓ Landsat TIR	×	× None	×
This Study (AquaVolt-AI DSI)	**10 m**	**Hourly**	**✓ Sentinel+ECOSTRESS+ERA5**	**✓ Explicit DSI**	**FAO-56 + PIML bounds**	**✓ Live**

The TSEB (Two-Source Energy Balance) model [14, 18] and the SEBAL (Surface Energy Balance Algorithm for Land) [3] are two of the most widely applied remote sensing energy balance models for ET retrieval. Both models rely on the canopy-air temperature difference as a primary driver of sensible heat flux estimation, and both are, by construction, sensitive to the confounded signal that decoupled stress imposes. A TSEB model run on a high-VPD, well-watered day will correctly retrieve a high latent heat flux from the soil surface (wet soil evaporation dominates) but may misattribute reduced canopy transpiration (due to stomatal closure) as a canopy water deficit, leading to overestimated water demand in the model's soil water balance module. The DSI framework introduced here provides the diagnostic tool needed to identify when such conditions occur and to flag model outputs for reinterpretation.

5.6. Implications for Remote Sensing-Based ET_c Products

Operational satellite-based evapotranspiration products, such as those derived from MODIS [6], ECOSTRESS, and Landsat thermal bands [2], are increasingly being used as the primary input to irrigation scheduling systems and water allocation policy at regional and national scales. These products derive ET_c by inverting the surface energy balance, and as such, their accuracy is directly coupled to the radiometric surface temperature (T_{rad}) retrieved from the satellite thermal instrument. When decoupled stress conditions prevail, the canopy radiometric temperature is elevated above what would be expected for a fully transpiring canopy, causing energy balance inversion algorithms to underestimate canopy latent heat flux (λE) and consequently overestimate sensible heat flux (H) and water stress.

This overestimation of crop water stress in operational ET_c products during high-VPD, adequate-soil-moisture conditions has direct policy consequences. In this pilot study, *all* 583 decoupled events occurred in corn. On corn-field high-VPD hours (VPD $>$ 2.0 kPa), decoupled events accounted for 461 of 8,640 high-VPD corn stress events (5.34%)—a small but non-zero fraction that would generate false positive irrigation triggers under a thermal-only scheduler operating on the corn field. This finding calls for the incorporation of VPD-based gating into satellite ET product generation pipelines, particularly for corn-type crops. The VPD data required for this correction is already available from ERA5-Land and MODIS atmospheric products at 10-km resolution.

5.7. Energy Balance Interpretation of Decoupled Stress Events

From a surface energy balance perspective, the decoupled stress condition can be understood as a partial transition between the potential evapotranspiration regime and the energy-limited regime, mediated by stomatal conductance. Under fully coupled conditions (adequate soil moisture, low VPD), the Penman–Monteith equation predicts that the canopy latent heat flux λE will approximately equal its potential value, and the Bowen ratio ($\beta = H/\lambda E$) remains low, typically below 0.3 for active vegetated canopies in summer conditions [11].

Under decoupled stress conditions, stomatal conductance (g_s) declines in response to high VPD, reducing transpiration below its potential rate. The energy that would have been dissipated as latent heat is instead repartitioned into sensible heat, raising canopy surface temperature and increasing the Bowen ratio. For the decoupled stress events identified in this study, we estimate the Bowen ratio shift as follows. The reduction in latent heat flux due to stomatal closure at VPD > 1.5 kPa can be estimated from the Penman–Monteith equation with a modified surface resistance r_s:

$$\lambda E_{decoupled} = \frac{\Delta\,(R_n - G) + \rho_a\, c_p\, \mathrm{VPD}\, g_a}{\Delta + \gamma\left(1 + \frac{g_a}{g_s^{stressed}}\right)} \tag{13}$$

where g_a is aerodynamic conductance (m/s), $g_s^{stressed}$ is stomatal conductance under VPD-induced closure (m/s), ρ_a is air density (kg/m^3), and c_p is specific heat of air (J kg^{-1} K^{-1}). When stomatal conductance halves from its reference value ($g_{s,ref} \approx 0.02$ m/s) due to VPD-induced closure, the latent heat flux under Sacramento Valley midday conditions (VPD = 2.0 kPa, R_n = 600 W/m^2) is reduced by approximately 35–45% from the potential rate, with the balance partitioned into sensible heat and canopy temperature rise of 3–5°C. This is fully consistent with the ΔT values observed in our decoupled stress events (mean ΔT = 3.8°C on high-VPD days).

This energy balance interpretation provides two important insights. First, it confirms that the thermal signal of decoupled stress is physically coherent and quantitatively predictable from atmospheric parameters alone, independent of soil moisture status. Second, it suggests that the magnitude of the thermal anomaly in a decoupled stress event provides information about stomatal conductance reduction rather than root-zone water depletion—a fundamentally different physiological variable with different agronomic management implications.

5.8. Pathways to Implementation in Smart Irrigation Controllers

The operationalization of the DSI framework in a smart irrigation controller requires three data streams that are already available in modern precision agriculture deployments:

1. **Real-time VPD**, derived from on-site air temperature and relative humidity sensors (typically available from a field weather station or portable IoT sensor nodes at costs of \$50–\$200 USD per unit).
2. **Root-zone depletion (D_r)**, either from a calibrated FAO-56 water balance model (computationally trivial) or from in-field capacitance soil moisture probes (readily available, cost approximately \$100–\$400 USD per probe).
3. **Canopy surface temperature (T_s)**, either from a handheld infrared thermometer (low-cost, field-deployable), a fixed canopy-level radiometer, or from satellite thermal revisits (Landsat every 8 days, ECOSTRESS 3–5 days depending on orbit).

The gating logic defined in Equation 12 requires only arithmetic operations on these three variables and can be implemented on a low-power microcontroller (such as an ESP32 or Arduino) at the edge of the field, enabling sub-minute response times without cloud connectivity. For deployments using satellite LST rather than field-level sensors, the DSI would be computed at each satellite overpass and stored as a farm management zone map, updated with each new thermal scene.

The water savings potential of DSI-gated irrigation scheduling is estimable from our real pilot data. Across the 21-day monitoring period, **583 sector-hours** of decoupled atmospheric stress were identified (all in corn). In a hypothetical thermal-anomaly-only scheduling system, each of these events might trigger an irrigation application of approximately 15–25 mm (a typical deficit-based application for California summer corn). Across 256 sectors (each representing 100 m^2 = 0.01 ha, total area = 2.56 ha), the total potential water saving is:

$$W_{saved} = N_{decoupled} \cdot \overline{I}_{applied} \cdot A_{sector} = 583 \times (15\text{–}25 \text{ mm}) \times 0.01 \text{ ha} \qquad (14)$$

This yields a real estimate of **87.5–145.8 mm total (34.2–57.0 mm/ha** over the pilot area), representing water that would be wasted by a thermal-only scheduler that does not gate on root-zone depletion status. This is consistent with efficiency ranges reported in deficit irrigation studies [5, 20].

5.9. Implications for Climate Change Adaptation in Mediterranean Agriculture

The Sacramento Valley, along with other Mediterranean-climate agricultural regions including the Murray–Darling Basin (Australia), the Guadalquivir Basin (Spain), and the Nile Delta (Egypt), is projected to experience significant increases in both mean temperature and VPD over the coming decades under high-emission climate scenarios [7]. Under Representative Concentration Pathway (RCP) 8.5, Sacramento Valley summer VPD is projected to increase by 0.3–0.8 kPa by 2050, which—based on the logistic regression model fitted in this study—would shift decoupled stress probabilities upward by 15–30 percentage points at the crop-canopy level. In practical terms, this means that as climate change intensifies, a growing fraction of crop thermal anomaly signals detected by satellite will reflect atmospheric demand rather than root-zone water depletion, making DSI gating increasingly critical for water-efficient irrigation management.

This has profound implications for the design of next-generation satellite agricultural monitoring systems, including the forthcoming ESA LSTM (Land Surface Temperature Monitoring) and NASA SBG (Surface Biology and Geology) thermal infrared missions. Both missions aim to provide diurnal LST coverage at 50–60-m resolution, significantly improving the temporal sampling of thermal stress signals compared to current Landsat and ECOSTRESS capabilities. The DSI framework presented in this study provides a ready-made validation methodology for these missions: by combining their high-temporal thermal observations with coincident VPD data and FAO-56 water balance model outputs, future studies can characterize the global prevalence of decoupled stress across diverse climate zones and crop types.

Furthermore, the coupling between VPD-induced decoupled stress and carbon flux dynamics deserves attention. When crops partially close stomata under high VPD, they simultaneously reduce CO_2 uptake, lowering canopy-level gross primary productivity (GPP) and net ecosystem exchange (NEE). This direct stomatal coupling between water and carbon fluxes [11, 19] means that decoupled stress events are not merely agronomic curiosities—they represent hours of reduced crop photosynthetic capacity with potential end-of-season yield consequences, even when root-zone water status remains adequate. Integrating the

DSI framework with solar-induced fluorescence (SIF) satellite products, which are sensitive to instantaneous photosynthetic rate, could provide a powerful new tool for simultaneously quantifying the water and carbon costs of decoupled stress events at field scale.

6. Conclusion

This study presents the first systematic quantification of decoupled crop water stress—the condition in which canopy thermal anomalies indicate stress while root-zone soil moisture remains within the readily available water threshold—at sub-field, 10-m spatial resolution using a 21-day real-time satellite-telemetry pilot deployment at the Russell Ranch Sustainable Agriculture Facility.

The key findings are as follows:

1. **Prevalence:** Of 72,094 stress-flagged sector-hour records across three active crop fields, 583 (0.80%) were classified as decoupled atmospheric stress—conditions that would generate false positive irrigation triggers in a thermal-only scheduling system. The dominant class was coupled root-zone stress (71,199 events; 97.84%), confirming that soil water depletion was the primary driver of crop thermal stress at Russell Ranch during summer 2026.
2. **Crop Dependency:** Decoupled stress was observed exclusively in the corn field (DSI = 0.0247). Alfalfa and tomato both yielded DSI = 0.000 over the monitoring period. The crop-type effect was highly significant ($\chi^2(6) = 3102.41$, $p < 0.001$).
3. **VPD Predictability:** Logistic regression on the corn field yielded $\beta = 0.823$, intercept $\alpha = -5.496$, AUC = 0.780. The model-implied VPD at 50% decoupled probability is 6.68 kPa—above the pilot monitoring range, but the model provides useful discrimination at lower probabilities. Alfalfa and tomato had zero decoupled events and were not modellable.
4. **Spatial Distribution:** Corn interior sectors showed higher mean DSI (0.0505) than border sectors (0.0146); however, this difference was not statistically significant ($t = -1.279$, $p = 0.206$) in the 21-day pilot. Sector-level ICC for corn was 0.2505, indicating meaningful between-sector variance. Alfalfa and tomato had zero sector-level variance (ICC = 0).

We introduce the Decoupled Stress Index (DSI) as a practical, computationally lightweight metric derived from standard FAO-56 water balance and satellite-derived LST data, and provide a gating logic formulation for operational irrigation scheduling. The pilot-scale estimate of potential false-irrigation water waste is **87.5–145.8 mm total** (34–57 mm/ha) over the 21-day corn-field observation window. This work demonstrates that simultaneous monitoring of atmospheric demand (VPD) and root-zone status (D_r) is essential for reliable, water-efficient irrigation scheduling.

Future work will extend this analysis to multi-season data, validate D_r estimates against direct soil moisture probes, expand spatial coverage, and integrate the DSI gating logic into the AquaVolt-AI operational dashboard for real-time scheduling applications.

Declarations

Ethical Approval and Consent

Not applicable. This study involves only remote sensing and computational analysis; no human subjects or animal experimentation were involved.

Consent for Publication

All authors consent to the publication of this manuscript.

Availability of Data and Materials

The AquaVolt-AI telemetry log and analysis code are available at `https://github.com/umertanveer25/aquavolt-ai-pk`. The Russell Ranch Century Experiment datasets are publicly available at `https://asi.ucdavis.edu/programs/ce/research/data`.

Competing Interests

The authors declare no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Authors' Contributions

U.T. conceived the study, designed the AquaVolt-AI pipeline, performed all data analysis, and wrote the manuscript. J.A. contributed to the agronomic interpretation and irrigation scheduling methodology. A.H. contributed to the statistical analysis framework. All authors reviewed and approved the final manuscript.

Acknowledgements

The authors are grateful to the Russell Ranch Sustainable Agriculture Facility (UC Davis) for providing open-access multi-decadal agronomic data. Sentinel-2 imagery was obtained from the Copernicus Open Access Hub. ERA5-Land reanalysis data was obtained from the Copernicus Climate Change Service (C3S) Climate Data Store.

References

[1] Allen, R.G., Pereira, L.S., Raes, D., Smith, M., 1998. FAO irrigation and drainage paper no. 56: Crop evapotranspiration. Food and Agriculture Organization of the United Nations 56, 1–300.

[2] Anderson, M.C., Norman, J.M., Kustas, W.P., Houborg, R., Starks, P.J., Agam, N., 2011. A thermal-based remote sensing technique for routine mapping of land-surface carbon, water and energy fluxes from field to regional scales. Remote Sensing of Environment 112, 4227–4241. doi:10.1016/j.rse.2007.07.018.

[3] Bastiaanssen, W.G.M., Menenti, M., Feddes, R.A., Holtslag, A.A.M., 1998. A remote sensing surface energy balance algorithm for land (SEBAL): 1. formulation. Journal of Hydrology 212-213, 198–212. doi:10.1016/S0022-1694(98)00253-4.

[4] Bellvert, J., Zarco-Tejada, P.J., Marsal, J., Girona, J., González-Dugo, V., Fereres, E., 2016. Airborne thermal imagery to detect the seasonal evolution of crop water status in peach, nectarine and saturn peach orchards. Remote Sensing 8, 39. doi:10.3390/rs8010039.

[5] Fereres, E., Soriano, M.A., 2007. Deficit irrigation for reducing agricultural water use. Journal of Experimental Botany 58, 147–159. doi:10.1093/jxb/erl165.

[6] Gao, F., Anderson, M.C., Kustas, W.P., Wang, Y., 2020. Mapping crop land-surface temperature from Landsat 8 and Sentinel-2. Remote Sensing of Environment 219, 1–14. doi:10.1016/j.rse.2018.09.030.

[7] Gerhards, M., Schlerf, M., Mallick, K., Udelhoven, T., 2019. Challenges and future perspectives of multi/hyperspectral thermal infrared remote sensing for crop water-stress detection. Remote Sensing 11, 1240. doi:10.3390/rs11101240.

[8] González-Dugo, V., Suárez, L., González-Piqueras, J., Zarco-Tejada, P.J., 2012. Thermal image indices to detect early water stress symptoms in maize. Agricultural Water Management 107, 103–113. doi:10.1016/j.agwat.2012.01.008.

[9] Idso, S.B., Jackson, R.D., Pinter, P.J., Reginato, R.J., Hatfield, J.L., 1981. Normalizing the stress-degree-day parameter for environmental variability. Agricultural Meteorology 24, 45–55. doi:10.1016/0002-1571(81)90032-7.

[10] Jones, H.G., 2004. Assessment of stomatal control of plant water status and the regulation of plant carbon gain from analysis of thermal images of leaves. Plant, Cell & Environment 27, 551–560. doi:10.1111/j.1365-3040.2004.01163.x.

[11] Katul, G.G., Oren, R., Manzoni, S., Higgins, C., Parlange, M.B., 2012. Evapotranspiration: a process driving mass transport and energy exchange in the soil-plant-atmosphere-climate system. Reviews of Geophysics 50. doi:10.1029/2011RG000366.

[12] Martínez-Ferrer, L., Moreno-Martínez, Á., Campos-Taberner, M., García-Haro, F.J., Piles, M., Adsuara, J.E., Muñoz-Marí, J., Camps-Valls, G., 2021. Quantifying uncertainty in high resolution biophysical variable retrieval with machine learning. Remote Sensing of Environment 280, 113199. doi:10.1016/j.rse.2022.113199.

[13] Moran, M.S., Clarke, T.R., Inoue, Y., Vidal, A., 1994. Estimating crop water deficit using the relation between surface-air temperature and spectral vegetation index. Remote Sensing of Environment 49, 246–263. doi:10.1016/0034-4257(94)90020-5.

[14] Norman, J.M., Kustas, W.P., Humes, K.S., 1995. Source approach for estimating soil and vegetation energy fluxes in observations of directional radiometric surface temper-

ature. Agricultural and Forest Meteorology 77, 263–293. doi:10.1016/0168-1923(95)02265-Y.

[15] Rafi, Z., Merlin, O., Le Dantec, V., Khabba, S., Er-Raki, S., Amazirh, A., 2023. Partitioning of evapotranspiration using high resolution thermal infrared imagery and TSEB model for wheat and tomato crops. Agricultural Water Management 275, 107989. doi:10.1016/j.agwat.2022.107989.

[16] Saliendra, N.Z., Sperry, J.S., Comstock, J.P., 1995. Influence of leaf water status on stomatal response to humidity, hydraulic conductance, and soil drought in Betula occidentalis. Planta 196, 357–366. doi:10.1007/BF00201396.

[17] Shen, C., et al., 2023. Differentiable modelling to unify machine learning and physical models for geosciences. Nature Reviews Earth & Environment 4, 552–567. doi:10.1038/s43017-023-00450-9.

[18] Timmermans, W.J., Kustas, W.P., Anderson, M.C., French, A.N., 2007. An intercomparison of the surface energy balance algorithm for land (SEBAL) and the two-source energy balance (TSEB) modeling schemes. Remote Sensing of Environment 108, 369–384. doi:10.1016/j.rse.2006.11.028.

[19] van der Tol, C., Verhoef, W., Timmermans, J., Verhoef, A., Su, Z., 2009. An integrated model of soil-canopy spectral radiances, photosynthesis, fluorescence, temperature and energy balance. Biogeosciences 6, 3109–3129. doi:10.5194/bg-6-3109-2009.

[20] Turner, N.C., 2004. Agronomic options for improving rainfall-use efficiency of crops in dryland farming systems. Journal of Experimental Botany 55, 2413–2425. doi:10.1093/jxb/erh154.

[21] Zhu, X., Dong, J., Dong, G., Chen, Y., 2024. Physics-informed deep learning for crop evapotranspiration estimation in data-scarce regions. Computers and Electronics in Agriculture 216, 108567. doi:10.1016/j.compag.2024.108567.